

VLSI Design of FIR Filter for DSP Applications

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Internship

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Project Overview

This project implements a 4-Tap FIR (Finite Impulse Response) Filter using Verilog HDL. FIR filters are widely used in Digital Signal Processing (DSP) applications such as noise reduction, audio processing, and communication systems.

Objectives

- Design a FIR filter using Verilog HDL.
- Implement filter coefficients using digital logic.
- Process digital input signals.
- Generate filtered output samples.

Features

- 4-Tap FIR Architecture
- Verilog HDL Implementation
- FPGA Compatible Design
- DSP Signal Processing

Filter Coefficients

Coefficient Set:

$$h(0) = 1$$

$$h(1) = 2$$

$$h(2) = 3$$

$$h(3) = 4$$

Files Included

- fir_filter.v
- fir_testbench.v
- README.md

- Project_Report.pdf

Applications

- Audio Processing
- Noise Filtering
- Image Processing
- Communication Systems
- Biomedical Signal Processing

Conclusion

The FIR filter successfully performs digital signal filtering using a 4-tap architecture and demonstrates the use of Verilog HDL in DSP applications.